

PROJECT REPORT
on
DIGITAL WATERMARKING

Submitted in partial fulfillment of requirements for the award of

Bachelor of Technology
in
Computer Science and Engineering



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FEB 2025



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CERTIFICATE

This is to certify that the project report entitled “*DIGITAL WATERMARKING*”, submitted to the School of Engineering and Technology (SET), THE ASSAM KAZIRANGA UNIVERSITY, JORHAT, ASSAM, in partial fulfillment for the degree of Bachelor of Technology in Computer Science and Engineering, is a record of Bonafide work carried out by Jyotiraditya Duarah - ET21BTHCS037 , Manash Dutta - ET21BTHCS050 , Samujjwal Das - ET21BTHCS047 and Mehboob Hasan - ET21BTHCS084 .

All help received from various sources have been duly acknowledged. No part of this report has been submitted elsewhere for award of any other degree.

ACKNOWLEDGEMENT

This project was jointly undertaken by Jyotiraditya Duarah , Manash Dutta , Samujjwal Das , and Mehboob Hasan . We would like to express our sincere gratitude to all those who have contributed to the successful completion of this project. This endeavor would not have been possible without the support, guidance and encouragement of various individuals and resources.

We express our gratitude to , for his continuous guidance and support. We also extend our thanks to all faculty members of the Department of Computer Science and Engineering for their insights and assistance throughout this project. Finally, we appreciate the resources provided by The Assam Kaziranga University , which contributed to the successful completion of this project.

We are grateful to all the faculty members of the Department of Computer Science and Engineering for providing a stimulating academic environment and for imparting knowledge that laid the foundation for this project.

Finally, we would like to thank our institution for providing us with the necessary resources and environment to pursue this project, fostering our growth and learning in the field of Computer Science.

ABSTRACT

Digital watermarking is a technique used to embed hidden information into digital media such as images, videos, audio, and documents for copyright protection, authentication, and security purposes. It plays a significant role in cybersecurity by ensuring data integrity, preventing unauthorized duplication, and detecting tampering.

In the digital era, safeguarding intellectual property, data integrity, and content authenticity is a major challenge. It helps organizations and content creators prevent unauthorized use, counterfeiting, and cyber threats.

Digital watermarking can be visible or invisible, depending on the application. Fragile watermarking detects alterations, while robust watermarking withstands modifications like compression and resizing, making it ideal for copyright enforcement and forensic tracking.

Its applications span media, cybersecurity, healthcare, forensics, and finance. It ensures data integrity, protects medical records, verifies digital evidence, and prevents leaks. As deep-fake technology and misinformation grow, advanced watermarking techniques, such as AI-driven and block-chain-based watermarking, are enhancing security and content traceability.

This project explores different types of digital watermarking techniques, their applications, and their role in protecting intellectual property.

CONTENTS

Certificate	i
Acknowledgement	ii
Abstract	iii
1.Introduction	6
1.1. Background	6
1.2. Purpose of the Project	6
1.3. Motivation	6
1.4. Problem Statement	7
1.5. Objectives	7
1.6. Scope of the Project	7
2. Literature Review	8
3. Project Details	
3.1 Overview	
3.2 Techniques Used	
3.3 Development Process	
3.3 Application Overview	
	9 - 11
4. Methodology and Approaches	12
5. Applications	14
6. Advantages and Limitations	16
7. Future Scopes	17
8. Conclusion	18
9. References	19

CHAPTER 1 : INTRODUCTION

1.1 BACKGROUND

In today's digital age, protecting intellectual property, ensuring data authenticity, and preventing unauthorized use of digital content have become significant challenges. The widespread availability of digital media, along with the rise of cyber threats and piracy, has increased the need for effective security measures to safeguard digital assets. Digital watermarking is a powerful technique that embeds hidden or visible information within digital content, such as images, videos, audio, and documents, to verify ownership, ensure authenticity, and detect tampering.

Unlike encryption, which restricts access to data, digital watermarking allows for tracking, verification, and authentication even after content is distributed. It can be classified into different types based on its visibility (visible or invisible) and robustness (fragile or robust). Visible watermarking is commonly used in branding and copyright enforcement, while invisible watermarking is useful for forensic tracking. Fragile watermarking is sensitive to alterations and helps detect tampering, whereas robust watermarking can withstand common modifications like compression and resizing, making it ideal for piracy prevention.

1. 2. PURPOSE OF THE PROJECT

Digital watermarking is a technique used to embed hidden information within digital content such as images, videos, audio, and documents. One of its primary purposes is copyright protection, ensuring that ownership details are embedded within the media to prevent unauthorized use. It also plays a crucial role in authentication and verification by proving the source of content and detecting any tampering. Additionally, digital watermarking helps with content tracking, allowing publishers and media companies to monitor the distribution and usage of their materials. In forensic analysis, watermarks assist in tracing leaks or unauthorized copies of sensitive content, making them valuable in legal investigations.

1.3. MOTIVATION

With the rise of cyber-crime, media piracy, and deep-fake technologies, there is an urgent need for improved authentication mechanisms. Digital watermarking provides a way to verify content integrity and protect intellectual property.

1.4. PROBLEM STATEMENT

With the rapid growth of digital content creation and distribution, protecting intellectual property, ensuring content authenticity, and preventing unauthorized use have become critical challenges. Traditional copyright mechanisms are often insufficient in deterring digital piracy, tampering, and illegal distribution. Digital watermarking offers a potential solution by embedding hidden or visible marks within digital media, allowing for ownership verification, tamper detection, and content tracking. However, existing watermarking techniques face challenges such as robustness against attacks (e.g., compression, cropping, or format conversion), maintaining media quality, and ensuring efficient extraction and verification. There is a need for an advanced, secure, and efficient digital watermarking system that can provide strong protection while maintaining the usability and integrity of digital content.

1.5. OBJECTIVES

- To study various types and techniques of digital watermarking.
- To explore the role of watermarking in cybersecurity and digital rights management.
- To analyze the effectiveness of watermarking techniques against cyber threats.
- To analyze the effectiveness of watermarking techniques against cyber threats.

1.6. SCOPE OF THE PROJECT

The Digital Watermarking project aims to develop a robust and efficient system for embedding, detecting, and verifying watermarks in digital media, including images, videos, audio, and documents. A key focus will be on ensuring security and robustness, making the watermark resistant to attacks like compression, scaling, cropping, format conversion, and noise addition while maintaining the quality of the media. Additionally, the project will provide content authentication by detecting unauthorized modifications and ensuring the integrity of digital files. It will also address ownership protection by embedding unique identifiers or metadata to assert copyright and prevent intellectual property theft. Furthermore, a user-friendly tool will be developed to enable users to apply, detect, and verify watermarks easily. The project will be implemented using programming languages such as Python or MATLAB, with a focus on balancing robustness, imperceptibility, and efficiency. The final outcome will be a secure and scalable digital watermarking system that effectively protects digital content while ensuring ease of use and accessibility.

CHAPTER 2 : LITERATURE REVIEW

Digital watermarking is a widely researched technique for embedding hidden information into digital media, such as images, videos, audio, and documents, to ensure copyright protection, authentication, and content integrity. The primary objective of digital watermarking is to provide security against unauthorized duplication, tampering, and distribution while maintaining imperceptibility and robustness.

TRADITIONAL WATERMARKING TECHNIQUES

The early digital watermarking techniques primarily focused on spatial domain embedding, where watermark data is directly modified within the pixel values of an image or the sample values of an audio signal. Common methods include the Least Significant Bit (LSB) technique, which modifies the least significant bits of pixel values to encode the watermark. While computationally efficient and easy to implement, spatial domain methods suffer from poor robustness, as they are highly susceptible to attacks such as compression, filtering, and geometric transformations.

HYBRID AND ADVANCED WATERMARKING TECHNIQUES

To overcome the limitations of individual watermarking approaches, hybrid methods combining multiple transforms have been developed. Techniques such as D WT-SV D and DC T-D WT-SVD leverage the advantages of multiple transformations, offering improved robustness, imperceptibility, and security. These hybrid approaches are particularly useful in multimedia applications where watermarking must withstand both intentional and unintentional distortions.

AI AND DEEP LEARNING-BASED WATERMARKING

The emergence of artificial intelligence, particularly deep learning, has significantly impacted watermarking research. Machine learning algorithms have been applied to improve watermark embedding and extraction processes. Convolutional Neural Networks (CNNs) and Generative Adversarial Networks (GANs) have been explored to create robust and imperceptible watermarking schemes. CNN-based models can learn complex patterns and adaptively embed watermarks, making them more resilient to attacks.

CHALLENGES AND FUTURE DIRECTIONS

Despite significant advancements, challenges remain in balancing watermark robustness, imperceptibility, and computational efficiency. Highly robust watermarking techniques often introduce perceptible distortions, while imperceptible techniques may be vulnerable to attacks. Future research aims to develop adaptive watermarking systems that dynamically adjust embedding strategies based on content characteristics and attack models. Additionally, integrating AI with Blockchain-based watermarking systems may provide enhanced security and real-time monitoring capabilities.

CHAPTER 3 : PROJECT DETAILS

3.1. OVERVIEW

The primary goal of this project is to develop a robust digital watermarking tool that enables users to embed, extract, and verify watermarks in various digital media formats. The tool will enhance copyright protection, authentication, and anti-piracy measures while maintaining media quality and security.

This project aims to contribute to digital rights management (DRM) by providing an advanced watermarking solution that ensures content authenticity, security, and traceability in the evolving digital landscape.

3.2. TECHNIQUES USED

1. Steganography for Watermarking

The project employs image steganography to embed invisible watermarks into images.

It modifies pixel values slightly to encode watermark information without significantly altering the visual appearance of the image.

The Least Significant Bit (LSB) technique is used to ensure that the watermark remains imperceptible.

2. Opacity & Color Control

Users can adjust the opacity and color of the watermark for better visibility.

A range slider allows fine-tuned control over opacity levels, modifying the alpha channel of the watermark image.

A color picker ensures users can choose custom watermark colors.

3. Drag-and-Drop File Upload

Users can upload images using an interactive drag-and-drop area. JavaScript detects when a file is dragged over the upload box and provides a visual highlight.

This improves the user experience compared to a traditional file input button.

4. Real-Time Image Preview

Uploaded images and watermarked results are displayed in real-time using JavaScript and CSS styling.

5. Download & Removal Functionality

After processing, users can download the watermarked image using a dynamically generated link.

A stylish download button with hover and scaling effects enhances interactivity.

Users can also remove uploaded images if necessary before applying a watermark.

3.3. DEVELOPMENT PROCESS

1. Frontend Development

The UI was built using HTML, CSS, and JavaScript.

CSS animations and transitions provide smooth effects when hovering over buttons and image elements.

The responsive design ensures compatibility across different devices, adapting elements for smaller screens.

2. Back-end Integration (Flask)

The back-end is developed using Python and Flask, allowing users to process images efficiently.

Flask handles file uploads, processes watermarking, and serves the modified images back to the front-end.

Steganography techniques and Open-CV functions are used for image processing.

3. Watermarking Algorithm Implementation

The watermarking algorithm applies a transparent watermark layer over the uploaded image.

The system supports both visible and invisible watermarking, depending on user preferences.

The LSB-based steganography algorithm is utilized for embedding hidden data into images.

4. Testing & Optimization

The tool was tested with different image formats (PNG, JPG) to ensure compatibility.

Optimizations included minimizing processing time, ensuring images remain high quality after watermarking.

Edge cases, such as handling large image files and unsupported formats, were addressed.

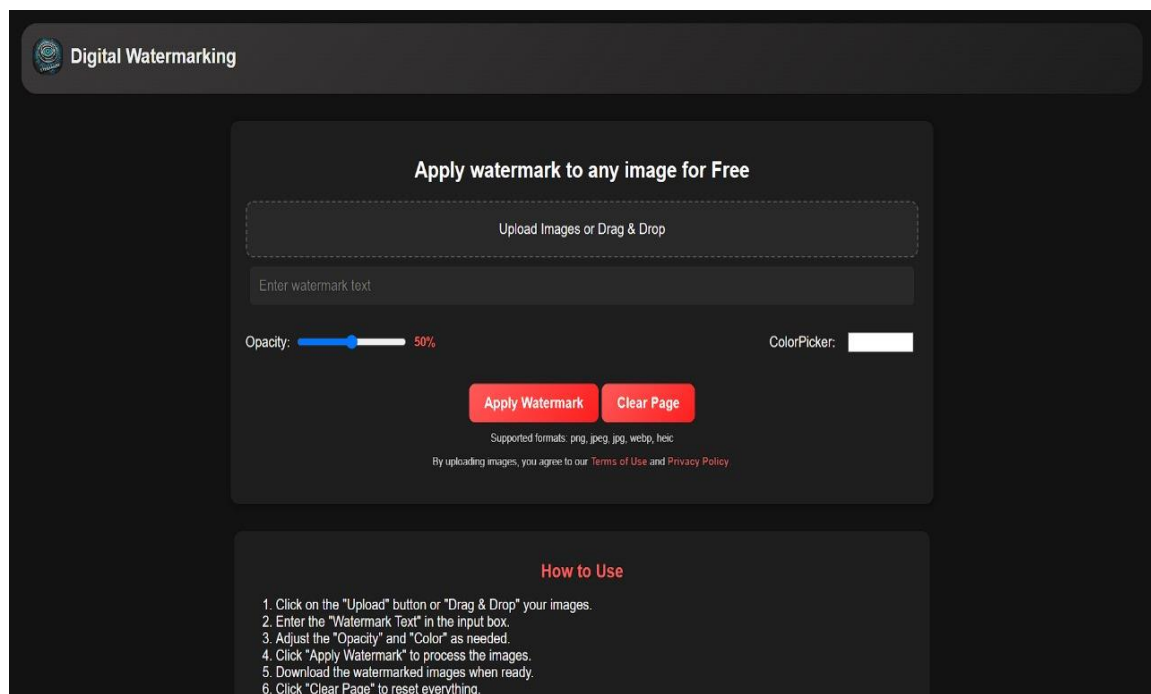
5. Final Enhancements

The UI was enhanced with an instructions panel for better usability.

Advanced color selection for watermarking was introduced.

A modern dark theme was applied for a professional and aesthetic look.

3.4. APPLICATION OVERVIEW



CHAPTER 4 : METHODOLOGY AND APPROACHES

1. TYPES OF WATERMARKING

- **VISIBLE WATERMARKING** - Visible Watermarking is a digital watermarking technique where a semi-transparent mark, such as a logo, text, or pattern, is embedded directly onto digital content like images, videos, or documents. Unlike invisible watermarking, which remains hidden, visible watermarks are intentionally placed to assert ownership, prevent unauthorized use, and discourage piracy.
- **INVISIBLE WATERMARKING** - Invisible Watermarking is a digital watermarking technique where an embedded mark is imperceptible to the human eye but can be detected using specialized software or algorithms. Unlike visible watermarking, which is clearly noticeable, invisible watermarking is designed to protect digital content without altering its appearance.
- **ROBUST WATERMARKING** - Robust Watermarking is a digital watermarking technique designed to withstand various forms of attacks and modifications, such as compression, scaling, cropping, noise addition, and format conversion. The primary goal of robust watermarking is to ensure that the embedded watermark remains intact and detectable even after the content undergoes transformations.
- **FRAGILE WATERMARKING** - Fragile Watermarking is a digital watermarking technique designed to detect any alterations or tampering in digital media. Unlike robust watermarking, fragile watermarks are highly sensitive to modifications, meaning that even slight changes to the content, such as compression, cropping, or noise addition, can alter or destroy the watermark.

2. IMPLEMENTATION TECHNIQUES

- **Least Significant Bit (LSB) Substitution** : LSB substitution is a simple and widely used technique for embedding watermarks in digital images. The idea is to replace the least significant bit (LSB) of each pixel with a watermark bit.
- **Discrete Cosine Transform (DCT)** : The Discrete Cosine Transform (DCT) is a frequency-domain technique used for watermarking. The DCT transforms the image into a frequency domain, where the watermark is embedded.
- **Discrete Wavelet Transform (DWT)** : The Discrete Wavelet Transform (DWT) is another frequency-domain technique used for watermarking. The DWT transforms the image into different frequency sub-bands, where the watermark is embedded.
- **Spread Spectrum Watermarking** : Spread spectrum watermarking is a technique that spreads the watermark across the entire image, making it more difficult to detect and remove.
- **Patchwork Watermarking** : Patchwork watermarking is a technique that embeds a watermark by modifying the statistical properties of the image.
- **Fractal Watermarking** : Fractal watermarking is a technique that uses fractal geometry to embed a watermark in an image.
- **Quantization Index Modulation (QIM)** : Quantization Index Modulation (QIM) is a technique that embeds a watermark by quantizing the image pixels.

CHAPTER 4 : APPLICATIONS

➤ COPYRIGHT PROTECTION

Digital watermarking is widely used for copyright protection by embedding ownership information into digital content such as images, videos, audio, and documents. This ensures that the rightful owner can claim authorship and prevent unauthorized use or distribution. If someone tries to copy or distribute the content without permission, the embedded watermark can serve as proof of ownership in legal disputes.

➤ CONTENT AUTHENTICATION AND INTEGRITY VERIFICATION

Watermarking is used to verify the authenticity and integrity of digital media. In fragile watermarking, any modification to the media (such as cropping, compression, or tampering) alters or removes the watermark, indicating potential unauthorized changes. This is crucial for applications where content integrity is essential, such as legal documents, financial records, and security-related communications.

➤ BROADCAST MONITORING

Media companies use digital watermarking to track their content across different platforms, including television, radio, and online streaming services. Watermarks embedded in broadcasted content help detect unauthorized use or distribution, ensuring that licensing agreements are followed. Monitoring agencies use specialized tools to scan and detect watermarks in various media outlets.

➤ DOCUMENT AUTHENTICATION AND ANTI-COUNTERFEITING

Official documents, certificates, and identity cards can be digitally watermarked to prevent forgery and counterfeiting. By embedding invisible watermarks in digital or scanned documents, organizations can verify their authenticity, reducing fraud in government-issued documents, academic certificates, and financial transactions.

➤ SECURE COMMUNICATIONS AND MILITARY APPLICATIONS

Digital watermarking is used in secure communications and military applications to verify the authenticity of classified documents, satellite images, and surveillance videos. By embedding secret watermarks, intelligence agencies can detect tampering and ensure secure information exchange.

- **SOCIAL MEDIA AND ONLINE CONTENT PROTECTION**
Watermarking is widely used in social media platforms to protect user-generated content, such as photographs, artwork, and videos. Many photographers, graphic designers, and content creators use visible watermarks to prevent unauthorized use of their work. Some platforms also use invisible watermarking to track content and prevent copyright infringement.
- **E-COMMERCE AND BRAND PROTECTION**
In e-commerce, watermarking helps protect brand identity by embedding watermarks in product images, preventing unauthorized sellers from misusing brand assets. This is particularly important for fashion brands, electronics, and digital advertisements, where counterfeiting is a major issue.
- **DIGITAL RIGHTS MANAGEMENT (DRM)**
Digital watermarking plays a key role in Digital Rights Management (DRM) systems, which control access and usage rights of digital content. For example, movies, music, e-books, and software can be watermarked to prevent unauthorized copying, distribution, and usage. If illegal copies are found, the watermark can help trace the source of the leak
- **FORENSIC TRACKING AND PIRACY DETECTION**
In forensic watermarking, unique, invisible watermarks are embedded in digital content before distribution. If the content is leaked or pirated, forensic analysis can help identify the original source of the leak. This is particularly useful for confidential documents, classified government data, and pre-release movies or software.
- **MEDICAL IMAGING SECURITY**
In the healthcare sector, digital watermarking ensures the authenticity and integrity of medical images such as X-rays, MRIs, and CT scans. Medical records often need to be transferred between hospitals, and watermarking helps prevent unauthorized alterations, ensuring that the diagnostic data remains accurate and secure. This is critical for patient safety and accurate medical analysis.

CHAPTER 5 : ADVANTAGES & LIMITATIONS

1. ADVANTAGES

- **Authentication & Integrity** : Watermarks serve as a digital signature that verifies the authenticity of media content. Ensures data authenticity and detects tampering.
- **Robustness** : A well-embedded watermark remains intact even if the media undergoes common processing, such as resizing or format conversion. This robustness ensures that ownership or authentication information is not easily removed or altered.
- **Imperceptibility** - A watermark is embedded in such a way that it does not degrade the quality of the original content.
- **Data Hiding** : Watermarks can embed metadata such as copyright information, author details, or usage restrictions without affecting content usability.
- **Tamper Detection** : If an image, video, or document is altered, the embedded watermark can reveal changes and confirm whether tampering has occurred.

2. LIMITATIONS

- **Vulnerability to Attacks** : Watermarks are susceptible to various types of attacks, including cropping, noise addition, or compression can weaken watermarks.
- **Computational Complexity** : Real-time watermarking for live video streaming or large-scale applications (e.g., cloud-based services) can be computationally expensive and time-consuming.
- **Capacity Constraints** : High-capacity watermarking may degrade visual or audio quality, while low-capacity watermarking may not store enough useful information. There is a limit to how much data can be embedded without causing noticeable distortions.
- **Removal or Collusion Attacks** : Attackers may attempt to remove or weaken watermarks using advanced techniques such as Filtering or Blurring, Machine Learning Attacks, Multiple Watermarking Conflicts.

CHAPTER 6 : FUTURE SCOPES

The future of digital watermarking is promising, with advancements in technology expanding its applications across various industries. One of the key developments is the integration of Artificial Intelligence (AI) and Machine Learning (ML) to enhance watermarking techniques, improving robustness, detection accuracy, and resilience against attacks. Additionally, Blockchain-based watermarking is expected to provide decentralized and tamper-proof content authentication, ensuring immutable proof of ownership and transparent tracking of digital assets. Another emerging area is quantum watermarking, which leverages quantum cryptography to create ultra-secure watermarking methods that are resistant to cyber threats.

With the rise of deep-fake technology, digital watermarking will play a crucial role in verifying the authenticity of images and videos, helping combat misinformation and media manipulation. Similarly, 5G networks and Internet of Things (IoT) security applications will benefit from watermarking, ensuring secure authentication and communication between connected devices. The growth of Augmented Reality (AR) and Virtual Reality (VR) also necessitates advanced watermarking techniques to protect digital assets in virtual environments, particularly in the meta-verse and immersive media platforms.

Furthermore, biometric and medical data security will see improvements through watermarking, preventing identity theft and unauthorized modifications of sensitive health information. The development of real-time and adaptive watermarking will enable dynamic embedding of watermarks into live-streamed content, safeguarding sports broadcasts, online lectures, and virtual meetings from piracy. Future watermarking techniques may also employ hybrid approaches, combining robust and fragile watermarking to provide layered protection while maintaining content quality.

CONCLUSION

Digital watermarking is a powerful technique for protecting digital content, ensuring authenticity, and preventing unauthorized use. It provides a secure way to embed ownership information, track distribution, and verify the integrity of digital media such as images, videos, audio, and documents. With applications in copyright protection, digital rights management, forensic tracking, medical imaging, and secure communications, watermarking plays a crucial role in various industries.

Despite its advantages, digital watermarking faces challenges such as robustness against attacks, maintaining media quality, and efficient extraction. Advancements in technology, including AI and machine learning, are helping to improve watermarking techniques, making them more secure and resistant to modifications. As digital content continues to grow exponentially, watermarking will remain a vital tool for content protection and cybersecurity.

In conclusion, digital watermarking is an essential technology for safeguarding digital assets in an era where content piracy, forgery, and unauthorized distribution are increasing. Future developments will further enhance its effectiveness, ensuring stronger protection and better adaptability across various digital platforms.

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